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Systems and methods for dynamic authorization of external devices for network access

Abstract

A system described herein may monitor information associated with a set of User Equipment (“UEs”) associated with a first network, and may receive an access request, for access to a first UE of the set of UEs, from a second UE associated with a second network. The system may obtain, based on the access request and from the second network, monitored information associated with the second UE. The system may identify monitored information associated with the first UE and a particular access policy that is associated with the first UE and the second UE. The system may identify, based on the particular access policy, the monitored information associated with the first UE, and the monitored information associated with the second UE, whether to grant or deny the access request. The system may output, in response to the access request, an indication of whether the access request is granted or denied.

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Background/Summary

BACKGROUND

(1) Wireless networks may provide services to User Equipment (“UEs”), such as mobile telephones, Internet of Things (“IoT”) devices, or other wireless devices. Different wireless networks may be owned and/or operated by different entities, or operators. Additionally, UEs may be registered, provisioned, etc. with certain networks that may perform functions such as authenticating UEs using network credentials (e.g., as maintained by a SIM (“Subscriber Identification Module”) card by UEs), maintaining usage or charging information, or the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an example overview of one or more embodiments described herein;
- (2) FIG. 2 illustrates an example of UE information determination by two separate networks, in accordance with some embodiments;
- (3) FIG. 3 illustrates an example of a dynamic authentication of an external UE for access to a network, in accordance with some embodiments;
- (4) FIG. 4 illustrates an example set of access policies that may be used in a dynamic authentication procedure, in accordance with some embodiments;

- (5) FIG. 5 illustrates an example process for a dynamic authentication of an external UE for access to a network, in accordance with some embodiments;
- (6) FIG. 6 illustrates an example environment in which one or more embodiments, described herein, may be implemented;
- (7) FIG. 7 illustrates an example arrangement of a radio access network (“RAN”), in accordance with some embodiments; and
- (8) FIG. 8 illustrates example components of one or more devices, in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

- (9) The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.
- (10) Embodiments described herein provide for intercommunication between different wireless networks in order to authenticate UEs that are associated with different networks (e.g., UEs that are associated with different “home” networks). A “home” network with respect to a given UE, as discussed herein, may refer to a wireless network that provisions network authentication credentials for the UE, that maintains charging or subscription service information for the UE, and/or performs other suitable operations with respect to the UE. Embodiments described herein provide for a first wireless network to maintain access policies associated with UEs that are associated with a second network as their respective home networks. Further, the access policies may include policies associated with UEs associated with the first network as their respective home networks. For example, the access policies may include criteria, conditions, etc. related to UEs of both the first and second networks, such as location, device integrity or operating status, security risk measurement scores, UE roles, and/or other parameters. In this manner, one network may be able to leverage authentication mechanisms, authorization mechanisms, location determination, device status, and/or other techniques implemented by another as well as such techniques implemented by the network itself, in order to provide for dynamic access control to the network.
- (11) For example, as shown in FIG. 1, UE **101-1** may be associated with a first network **103-1**, and UE **101-2** may be associated with a second network **103-2**. For example, network **103-1** may be a home network of UE **101-1**, and network **103-2** may be a home network of UE **101-2**. Networks **103-1** and **103-2** may each be, or may each include, a mobile network, a public network, a private network, a wireless network, etc.
- (12) In some embodiments, networks **103-1** and **103-2** may be different types of networks and/or may otherwise have differing attributes. For example, network **103-1** may be a mobile network (e.g., an Long-Term Evolution (“LTE”) network, a Fifth Generation (“5G”) network, or the like), and network **103-2** may be a private network. The private network may include one or more gateways, firewalls, routers, etc. and/or other mechanisms to restrict access or visibility to unauthorized devices. In some embodiments, the private network may be implemented by, and/or may utilize, some of the same resources as network **103-1**. For example, network **103-1** may include wireless network infrastructure resources such as base stations, radios, antennas, RAN controllers, or the like, and may allocate portions of these resources for use by network **103-2**. For example, network **103-1** may allocate a portion of radio frequency (“RF”) bandwidth, spectrum, etc. for use by network **103-2**. In some embodiments, network **103-2** may be separate from network **103-1**, inasmuch as networks **103-1** and **103-2** may utilize no common resources. For example, network **103-2** may include a private wireless local area network (“WLAN”) implemented by WiFi access points, routers, etc.
- (13) As discussed herein, networks **103-1** and **103-2** (sometimes referred to in plural as “networks **103**”) may monitor or maintain authentication information, authorization information, location information, device integrity information, security risk measurement score, role associate with the device, and/or other suitable information associated with UEs **101-1** and **101-2** (sometimes referred to in plural as “UEs **101**”), respectively. For example, as shown in FIG. 2, networks **103** may

include one or more network elements or other devices or systems that determine, monitor, and/or maintain such information.

(14) For example, network **103-1** may include authentication and authorization system **201-1** (referred to herein as “authentication system” for the sake of brevity), device integrity system **203-1**, location determination system **205-1**, security risk measurement system **209-1**, and role-based directory system **211-1**; and network **103-2** may include authentication system **201-2**, device integrity system **203-2**, location determination system **205-2**, security risk measurement system **209-2**, and role-based directory system **211-2**. As similarly noted above, authentication system **201**, device integrity system **203**, and location determination system **205** may include and/or may be implemented by different types or arrangements of devices or systems with respect to different networks **103**. For example, network **103-1** may utilize a first type of authentication system **201-1** that uses a first type of authentication mechanism to authenticate UEs (e.g., UE **101-1**), while network **103-2** utilizes a second type of authentication system **201-2** that uses a different second type of authentication mechanism to authenticate UEs (e.g., UE **101-2**). Further, in some embodiments, authentication system **201**, device integrity system **203**, location determination system **205**, security risk measurement system **209**, and/or role-based directory system **211** of different networks **103** may perform different sets of functionality, such as some or all of the operations discussed below, or a portion thereof. For example, authentication system **201-2** of network **103-2** may communicate with elements of network **103-1**, as discussed below, but authentication system **201-1** of network **103-1** may not be configured to communicate with elements of network **103-2** in a similar manner. On the other hand, authentication system **201**, device integrity system **203**, location determination system **205**, security risk measurement system **209**, and/or role-based directory system **211** of different networks **103** may perform identical or similar functionality with respect to UEs **101** of the different networks **103**.

(15) As shown, authentication system **201** may authenticate respective UEs **101**, such as by verifying that credentials (e.g., a Subscription Permanent Identifier (“SUPI”), a Globally Unique Temporary Identifier (“GUTI”), and/or some other suitable credentials or values) provided by UEs **101** are accurate and/or otherwise satisfy authentication procedures implemented by network **103**. Additionally, or alternatively, authentication system **201** may authenticate respective UEs **101** by verifying a user name, password, biometric information, cryptographic keys and/or values derived based on such keys, an authentication token, an authorization token (e.g., an OAuth token, a Security Assertion Markup Language (“SAML”) assertion, etc.), and/or other suitable authentication and/or authorization information.

(16) In some embodiments, UEs **101** may include a SIM card, a Universal Integrated Circuit Card (“UICC”), an embedded SIM (“eSIM”), a key repository (e.g., storing one or more keys such as public/private key pairs), and/or some other authentication mechanism by which UEs **101** may communicate with authentication system **201** to perform an authentication procedure. In some embodiments, authentication system **201** may include an Authentication, Authorization, Accounting (“AAA”) system, an Authentication Server Function (“AUSF”), a proprietary or non-standard authentication system, etc. Authentication system **201** may maintain and/or provide authentication and/or authorization information associated with UE **101-1**, such as an indication of when UE **101-1** was authenticated by authentication system **201-1** (e.g., a timestamp), a type of authentication performed (e.g., verification of SIM credentials, verification of a user name or password, verification of biometric information, etc.), and/or other suitable information. Authorization information may include the resources that a UE **101-1** is allowed to access, the duration of access, expiration of the authorization information and one or more granular services or resources that the UE **101-1** is authorized for, also sometimes referred to as “scope.”

(17) Device integrity system **203** may monitor, determine, etc. a measure of device integrity, operational status, device health, and/or other suitable measures (referred to herein simply as “device integrity” for the sake of brevity). Device integrity of UE **101** may include and/or may be

based on, for example, a determination that UE **101** is operational, powered on, has connectivity, etc. Such status may be determined based on, for instance, “ping” messages sent to UE **101** and responses thereto, “heartbeat” messages periodically and/or intermittently sent by UE **101** to device integrity system **203**, and/or other suitable techniques. In some embodiments, device integrity may include an indication of whether UE **101** is in a “lost” mode, such as a reduced-functionality mode in which only emergency calls are able to be made and/or in which additional measures are required to be taken (e.g., in addition to a “screen unlock” function) in order to access or “unlock” UE **101**.

(18) As another example, device integrity may include a verification that UE **101** has a particular operating system, firmware version, application or application suite, or other set of software installed or executing at UE **101**. In some embodiments, device integrity may include a measurement of one or more values associated with the various components of the UE **101** (e.g., software, firmware, hardware, configuration files, etc.) that indicate that the components are operating according within acceptable parameters and that an attacker has not tampered with any of the components. In some embodiments, device integrity may include a verification of device health parameters associated with UE **101**, such as at least a threshold battery level (e.g., at least 50% charged, at least 80% charged, etc.), at least a threshold amount of processing and/or memory resources available, no water immersion flags, etc. In some embodiments, device integrity may include or may be based on sensor information measured by one or more sensors of UE **101** such as an accelerometer (e.g., which may indicate drops, impacts, or other damage to UE **101**), a thermometer (e.g., which may indicate UE **101** overheating), a camera, and/or other sensors or input devices of UE **101**. In some embodiments, device integrity information may include other suitable types of information that may be collected, maintained, provided, etc. by UE **101**. In some embodiments, UE **101** may communicate with device integrity system **203** via an application programming interface (“API”), an application executing at UE **101** (e.g., which is configured to communicate with device integrity system **203**), a web portal associated with device integrity system **203**, and/or some other suitable communication pathway.

(19) Location determination system **205** may monitor, determine, receive, etc. location information associated with UE **101**. For example, location determination system **205** may include, may be implemented by, and/or may be communicatively coupled to, a mobility management system of network **103**, such as a Mobility Management Entity (“MME”), an Access and Mobility Management Function (“AMF”), or other suitable device or system of network **103**. In some embodiments, location determination system **205** may communicate with base stations of a RAN associated with network **103** in order to determine the location (e.g., using triangulation or other suitable techniques) of UE **101**. Additionally, or alternatively, location determination system **205** may receive location information from UE **101**, which may be determined by UE **101** using geographical Global Positioning System (“GPS”) techniques, geofencing techniques, and/or other suitable techniques.

(20) Security risk measurement system **209** may include components that determine or indicate a measure of potential risk that UE **101** poses to the network and the data within the network. Security risk measurement system **209** may include components that check for security vulnerabilities (e.g., vulnerable coding such a buffer overflow), lack of system hardening, and penetration testing of the UE **101**. The measure of potential risk may additionally, or alternatively, be based on static application security testing (“SAST”) performed with respect to UE **101** and/or dynamic application security testing (“DAST”) of UE **101**. The results from the various security testing may be used based (e.g., as a weighted average) for a consolidated security risk measurement value for a UE **101**.

(21) Role-based directory system **211** may make use of database or a list of roles or groups to which the UE **101** may belong. Roles may be pre-assigned based on certain characteristics (e.g., type of UE **101**, application running on the UE **101**, administrator operating the UE **101**, etc.) of a

given UE **101**. The roles of such UE **101** may be re-assigned if one or more characteristics changes or if the policy changes are made within the network.

(22) In some embodiments, a particular device or system of each network **103**, such as Unified Data Repository (“UDR”) **207-1** of network **103-1** or UDR **207-2** of network **103-2**, may receive some or all of the information (e.g., authentication and/or authorization information, device integrity information, location information, security risk measurement information, role/group information, etc.) associated with respective UEs **101**. In some embodiments, some other device or system of network **103-1** and/or network **103-2** may maintain the authentication information, device integrity information, location information, security risk measurement information, and role/group information.

(23) Returning to FIG. **1**, UE **101-1** may request (at **102**) access to UE **101-2**. As discussed above, network **103-1** may be a home network of UE **101-1**, and network **103-2** may be a home network of UE **101-2**. The access request (at **102**) may, for example, be sent from UE **101-1** to UE **101-2** (e.g., using an Internet Protocol (“IP”) address, Uniform Resource Locator (“URL”), Mobile Directory Number (“MDN”), Subscription Permanent Identifier (“SUPI”), International Mobile Subscriber Identity (“IMSI”), a temporary identifier, and/or other identifier of UE **101-2**). In this sense, UE **101-1** may be referred to herein as a “requesting UE,” and UE **101-2** may be referred to as a “target UE” (e.g., for which access is being requested).

(24) The request may be sent via network **103-1**, network **103-2**, and/or some other network (e.g., the Internet). In some embodiments, the request may be sent by a device other than UE **101-1** (e.g., a workstation, a laptop, a wearable device, another UE, etc.), and may indicate that UE **101-1** is requesting access to receive a particular service and/or otherwise communicate with UE **101-2**. In some embodiments, the request may be received by UE **101-2** and may be forwarded by UE **101-2** to authentication system **201-2**. Additionally, or alternatively, the request may be sent to authentication system **201-2**, and/or some other network element of network **103-2** may receive the request and forward the request to authentication system **201-2**.

(25) As described in more detail below, network **103-2** (e.g., authentication system **201-2** of network **103-2**) may obtain (at **104**) information associated with UE **101-1**, such as location information, device integrity information, authentication and/or authorization information, security risk measurement information, role/group information, and/or other suitable information, from the home network of UE **101-1** (i.e., network **103-1**, in this example). Based on obtaining (at **104**) the information regarding UE **101-1** from network **103-1**, and further based on location information, device integrity information, authentication and/or authorization information, security risk measurement information, role/group information, and/or other suitable information associated with UE **101-2**, network **103-2** may determine (at **106**) whether to grant the requested access. For example, as discussed below, network **103-2** may maintain access policies including conditions, parameters, etc. relating to location information, device integrity information, authentication and/or authorization information, security risk measurement information, role/group information, and/or other suitable information associated with UEs that are associated with network **103-2** as a home network (e.g., UE **101-2**), as well as UEs that are not associated with network **103-2** as a home network (e.g., UE **101-1**). In this sense, status changes, configuration changes, security risk measurement value changes, role/group information changes, location changes, and/or other dynamic information with respect to UEs that have differing networks as their home networks, may be able to be readily accounted for and used to dynamically determine whether to grant access to UEs of other networks to communicate with UEs associated with network **103-2**.

(26) Accordingly, network **103-2** may respond (at **108**) to the access request, such as by indicating to UE **101-1** that the requested access has been granted, has been granted with certain parameters or restrictions (e.g., a time limit, a restriction on types of traffic or services, etc.), or has been denied. In some embodiments, network **103-2** may perform one or more other actions, such as notifying UE **101-2** that UE **101-1** has been granted access to communicate with UE **101-2**. In such

situations, UEs **101-1** and **101-2** may proceed to communicate (at **110**) with each other in order to provide the requested service or other type of communication.

(27) As shown in FIG. 3, example access request **301** may be received (at **102**) by authentication system **201-2** of network **103-2**. In some embodiments, access request **301** may include an identifier of the requesting UE. In this example, the identifier may include an identifier of UE **101-1**, such as an International Mobile Station Equipment Identity (“IMEI”) value, a Generic Public Subscription Identifier (“GPSI”), a SUPI value, an IMSI value, an MDN, an IP address, or other suitable identifier of UE **101-1**. Generally, the UE identifier may be an identifier that may be used by the home network of UE **101-1** to identify records, authentication and/or authorization information, or other information associated with UE **101-1**.

(28) In some embodiments, access request **301** may include an identifier of a home network of UE **101-1**, such as a Public Land Mobile Network (“PLMN”) identifier, an IP address (e.g., an IP address, or other locator information of a gateway, interface, etc. of network **103-1**), a realm name, a domain name, a carrier name, or other suitable network identifier. Generally, the network identifier may be used by authentication system **201-2** to identify which particular network **103** to communicate with in order to obtain the location information, device integrity information, authentication information, security risk measurement information, role/group information, and/or other suitable information associated with UE **101-1**. In this sense, network **103-2** may be considered a “roaming” network with respect to UE **101-1**, a “non-native” network with respect to UE **101-1**, a “non-home” network with respect to UE **101-1**, etc. Additionally, or alternatively, UE **101-1** may be considered a “roaming,” “external,” “non-native,” etc. UE with respect to network **103-2**, and/or otherwise may be considered a UE for which network **103-2** is not a home network. As noted above, network **103-2** may be a private network and/or a different type of network than network **103-1**, and may utilize different techniques, schemes, methodologies, etc. for UE authentication and/or authorization, device integrity verification, security risk measurement information, role/group information, UE location monitoring, etc.

(29) Access request **301** may further include a service type. A service type may include, for example, an identifier or descriptor of a category of services, such as “automated guided vehicle (“AGV”) control,” “web browsing,” “content streaming,” “sensor monitoring,” and/or other categories. In some embodiments, a service type may include an identifier or descriptor of a particular application, such as an application name, an application identifier, etc.

(30) Access request **301** may additionally, or alternatively, include an identifier of a target UE (i.e., UE **101-2**, in this example). For example, the identifier of the target UE may include an IP address, SUPI, temporary identifier (e.g., GUTI, 5G-GUTI, Temporary Mobile Subscriber Identity (“TMSI”), etc.) or other identifier of UE **101-2**. The IP address or other identifier may, in some embodiments, be specifically indicated in the request (e.g., may be specified by the requesting UE **101-1**). Additionally, or alternatively, the IP address or other identifier may be provided by one or more routing elements of network **103-2**, which may have performed network address translation (“NAT”), mapping, and/or other suitable operations to translate, convert, etc. an identifier provided by requesting UE **101-1** to the IP address or other identifier of UE **101-2** within network **103-2**. The identifier of the target UE may include some other identifier of UE **101-2** in addition to or in lieu of an IP address, such as an MDN, a device name, and/or some other suitable identifier. In some situations, a temporary identifier (e.g., a 5G-GUTI) may be mapped to a SUPI, such that privacy of UE **101-2** is not compromised to elements external to network **103-2**, and preventing physical tracking of UE **101-2**.

(31) In some embodiments, authentication system **201-2** and/or some other element of network **103-2** may maintain mapping information that associates UE **101-2** and/or a group of UEs (e.g., including UE **101-2**) to a particular service type. For example, access request **301** may indicate a service type but may not include an identifier of UE **101-2**. In such examples, authentication system **201-2**, a routing element of network **103-2**, a load balancer, and/or some other device or

system may determine (e.g., based on the mapping information) that UE **101-2** is associated with the requested service type. As another example, access request **301** may indicate an identifier of UE **101-2** but may not include an indication of the requested service type. In such examples, authentication system **201-2**, a routing element of network **103-2**, a load balancer, and/or some other device or system may determine (e.g., based on the mapping information) that access request **301**, directed toward UE **101-2**, is a request for a particular service type.

(32) Based on receiving (at **102**) access request **301**, authentication system **201-2** may identify that the requesting UE **101-1** is associated with network **103-1** (e.g., as a home network of UE **101-1**). As shown in FIG. 3, authentication system **201-2** may register (at **302**) with network **103-1**. For example, authentication system **201-2** may register (at **302**) with network **103-1** based on receiving (at **102**) access request **301**, and/or may have registered with network **103-1** at some prior time. The registration (at **302**) with network **103-1** may include registering with an network information exposure element of network **103-1**, such as Network Exposure Function (“NEF”) **303**, a Service Capability Exposure Function (“SCEF”), and/or some other suitable device or system. NEF **303** may, for example, perform authentication and authorization functions in order to authenticate information requests from authentication system **201-2**, and to provide types or amounts of information that authentication system **201-2** is authorized to receive. That is, NEF **303** may, in some embodiments, maintain information indicating which devices or systems are authorized to receive different types of information. In this manner, different networks **103** and/or elements thereof (e.g., respective authentication systems **201** and/or other network elements) may be authorized to receive different types of information, such as information associated with different groups of UEs **101** that are associated with network **103-1** as a home network.

(33) For example, as part of the registration (at **302**), NEF **303** may maintain information indicating that authentication system **201-2** is authorized to access location information associated with some or all UEs **101** registered with network **103-1** as a home network, device integrity information associated with some or all UEs **101** registered with network **103-1** as a home network, and authentication information associated with some or all UEs **101** registered with network **103-1** as a home network. In some embodiments, types of information that are not registered (at **302**) as being authorized for authentication system **201-2** to access may not be available to authentication system **201-2**. For example, in some situations, NEF **303** may receive or maintain other information, such as usage or billing information associated with one or more UEs **101**. In the event that authentication system **201-2** has not been registered (at **302**) as being authorized to access such information, authentication system **201-2** may not be able to access this information (e.g., requests for such information may be denied).

(34) In some embodiments, the authorization information, maintained by NEF **303** with respect to authentication system **201-2**, may indicate parameters, constraints, characteristics, etc. of access. For example, NEF **303** may maintain information indicating that authentication system **201-2** is authorized to receive the last one hour of location history with respect to a particular set of UEs **101**, but that another authentication system **201** (e.g., associated with a different network **103**) is authorized to receive the last two hours of location history with respect to the same particular set of UEs **101**. As another example, NEF **303** may maintain information indicating that authentication system **201-2** is authorized to request the initiation of an authentication procedure with respect to one or more UEs **101**, but that another authentication system **201** (e.g., associated with a different network **103**) is not authorized to request the initiation of an authentication procedure with respect to the same one or more UEs **101**. For instance, authentication system **201-2** may be authorized to request that one or more elements of network **103-1** (e.g., authentication system **201-2**) perform an authentication procedure to verify credentials of UE **101-1** in response to the request from authentication system **201-2**, whereas the other authentication system **201** may only be authorized to receive results of a previously performed authentication (e.g., credential verification) of UE **101-1**. As another example, authentication system **201-2** may be authorized to perform an

authentication and authorization procedure to verify credentials of UE **101-1** using secondary credentials that may be provided by or associated with network **103-2**, and are used by UE **101-1**. The authentication and authorization procedures may be performed using a secondary authentication mechanism or network slice-specific authentication and authorization procedures. In such a scenario, UE **101-1** may have an existing or prior relationship with network **103-2**.

(35) As another example, NEF **303** may maintain information indicating different parameters of authorized information based on different service types. For example, NEF **303** may maintain information indicating that authentication system **201-2** is authorized to receive one hour of UE location history for a “AGV control” service, but is only authorized to receive five minutes of UE location history for a “content streaming” service.

(36) Based on identifying that the requesting UE **101-1** is associated with network **103-1**, with which authentication system **201-2** has previously registered (at **302**), authentication system **201-2** may request (at **304**) information associated with UE **101-1**. The request (at **304**) may include the identifier of UE **101-1** included in access request **301**. In some embodiments, the request (at **304**) may include other information, such as an identifier of network **103-2**, the requested service type, and/or an identifier of the target UE **101-2**. NEF **303** may determine, based on the registration (at **302**) of authentication system **201-2** that authentication system **201-2** is authorized to receive the requested information, or at least a portion of the requested information (e.g., is authorized to receive some but not all location information associated with UE **101-1**, or in some instances authentication system **201-2** is authorized to receive all available location information associated with UE **101-1**).

(37) In some embodiments, the request (at **304**) may include an identifier of network **103-2** or of authentication system **201-2** and an identifier of UE **101-1**, but may not include the service type and/or the identifier of the target UE **101-2**. In such situations, NEF **303** may determine which types of information, and/or parameters of such information, that authentication system **201-2** is authorized to access based on the previous registration (at **302**) of authentication system **201-2**. For example, as noted above, different networks **103** and/or authentication systems **201** may be authorized to access different types of information. In some situations, the requested (at **304**) information may exceed the types or amounts of information that authentication system **201-2** is authorized to receive. In such situations, NEF **303** may identify, out of the requested information, which information authentication system **201-2** is authorized to receive.

(38) In some embodiments, in addition to the operations shown in FIG. 3, one or more elements of network **103-1** may perform one or more additional operations related to the request (at **304**). For example, NEF **303** may provide some or all of the information included in the request to authentication system **201-1**, which may also authenticate UE **101-1** and/or verify that UE **101-1** is authorized to receive the requested service type. For example, authentication system **201-1** may identify one or more policies maintained in UDR **207-1** and/or some other device or system (e.g., a PCF, a PCRF, etc.) of network **103-1**. In instances where network **103-1** (e.g., authentication system **201-1**) determines that UE **101-1** is not authenticated, not able to be authenticated, is not authorized to receive the requested service, and/or that the request should otherwise not be granted, authentication system **201-1** may respond (e.g., via NEF **303**) to the request (at **304**) with an indication that the request should be denied (e.g., the requested access should not be provided). In this manner, policies associated with the home network **103-1** of UE **101-1** may be able to be accounted for and adhered to in the authorization of UE **101-1** for access to network **103-2**, which is not a home network of UE **103-1**.

(39) Assuming, on the other hand, that UE **101-1** is authorized to receive the requested service, and further assuming that authentication system **201-2** is authorized to receive the requested UE information (e.g., that UE **101-1** is not prohibited or restricted from receiving the requested service, and/or that authentication system **201-2** is not prohibited or restricted from receiving the requested UE information), NEF **303** may obtain (at **306**) the UE information, which authentication system

201-2 is authorized to receive, from **UDR 207-1** and/or some other suitable device or system. In this example, **NEF 303** may obtain (at **306**) location information, device integrity information, and authentication and/or authorization information associated with **UE 101-1**. In some embodiments, **NEF 303** may obtain (at **306**) other information, such as security risk measurement information, role/group information, etc. In some situations, as noted above, authentication system **201-2** may be authorized to request (e.g., at **304**) new or “fresh” information (e.g., location information determined within the last 30 seconds, a verification of authentication of **UE 101-1** performed after the request (at **304**), a verification of device integrity of **UE 101-1** performed after the request, etc.). In some situations, **NEF 303** and/or **UDR 207-1** may communicate with authentication system **201-1**, device integrity system **203-1**, location determination system **205-1**, security risk measurement system **209-1**, and/or role-based directory system **211-1** in order to obtain the new or “fresh” information (e.g., where authentication system **201-1**, device integrity system **203-1**, location determination system **205-1**, security risk measurement system **209-1**, and/or role-based directory system **211-1** communicates with **UE 101-1** and/or otherwise respectively determines the requested information).

(40) **NEF 303** may provide (at **308**) the requested information to authentication system **201-2**. Authentication system **201-2** may also obtain (at **310**) information associated with target **UE 101-2**, such as location information, device integrity information, and authentication information. For example, authentication system **201-2** may obtain (at **310**) such information from **UDR 207-2** and/or some other suitable device or system.

(41) Authentication system **201-2** may also identify (at **312**) an access policy associated with the request. For example, network **103-2** may include policy repository **305**, which may be, may include, may be implemented by, may be communicatively coupled to, and/or may be otherwise associated with a Policy Control Function (“PCF”), a Policy Charging and Rules Function (“PCRF”), and/or some other suitable device or system. Policy repository **305** may, for example, may maintain one or more access policies based on information associated with **UEs 101** that are associated with network **103-2** as a home network as well as with **UEs 101** that are associated with one or more other networks (e.g., network **103-1**).

(42) For example, as shown in FIG. 4, policy repository **305** may maintain data structure **401**, which may include an example set of access policies that may be used (e.g., at **314**) by authentication system **201-2** to respond to access requests (e.g., access request **301** from **UE 101-1** and/or other requests). In some embodiments, different access policies maintained by policy repository **305** may be associated with different particular requesting **UEs** or groups of requesting **UEs**, particular target **UEs** or groups of target **UEs**, and/or service types. For example, as shown, a first access policy may be associated with two particular requesting **UEs 101-1** and **101-2**, and may be associated with a particular target **UE 101-2**. This access policy may also be associated with the service type “AGV control.” Thus, authentication system **201-2** may identify that the example access request **301** matches the criteria of this access policy (denoted by the dashed box and italic lettering in FIG. 4), because the requesting **UE 101-1**, target **UE 101-2**, and service type indicated in access request **301** match the criteria indicated in data structure **401**. In some embodiments, authentication system **201-2** may perform some other type of suitable similarity analysis to identify the access policy that most closely matches or meets the parameters indicated in access request **301**.

(43) As further shown, access policies may include additional conditions or criteria, which may be based on authentication and/or authorization information, device integrity information, location information, and/or other information associated with the requesting **UE** and/or the target **UE**. For example, the identified access policy may indicate that requested access should be granted if a location of the requesting **UE** is within a first geographical region (represented as “Region_A”) and if a location of the target **UE** is within a second geographical region (represented as “Region_B”). The geographical regions may be specified in terms of latitude and longitude coordinates and/or

boundaries, city names, cell sector identifiers, physical addresses, and/or other suitable indicators of geographical location.

(44) Access policies may also specify limits or constraints on access if other conditions are met. For example, the identified access policy may specify that, assuming the other conditions are met (e.g., the requesting UE is within Region_A and the target UE is within Region_B), the requesting UE is authorized to access (e.g., receive service from, communicate with, etc.) the target UE for two hours. After the two hours elapse, the authorization for the requesting UE to access the target UE may lapse, and the target UE may cease communicating with or providing service to the requesting UE, and/or one or more network elements of network **103-2** may cease forwarding traffic between the requesting UE and the target UE based on the expiration of this duration.

(45) As additionally shown in FIG. 4, another access policy may be applicable to requesting UE **101-1**, target UE **101-4**, and a “web browsing” service type. This access policy may specify that the authentication information associated with the requesting UE should be two days old or newer. For example, if authentication system **201-2** receives information indicating (e.g., from a home network of a requesting UE) that the requesting UE has been authenticated three days ago by the home network, then authentication system **201-2** may reject a request from this UE and/or may request that the home network provide a new or “fresh” authentication of the requesting UE. This access policy may further specify that access is granted for the requesting UE to access the target UE if the requesting UE and the target UE are within a particular distance of each other (i.e., 90 meters, in this example). For example, in some embodiments, authentication system **201-2** may periodically or intermittently request (e.g., via a “pull” mechanism) or receive (e.g., via a “push” mechanism) location information from a home network of the requesting UE, and/or may periodically or intermittently monitor location information (e.g., as maintained by UDR **207-2**) of the target UE to verify that such UEs remain within 90 meters of each other. In situations where access has previously been granted but conditions change (e.g., where the requesting UE and/or the target UE move such that these UEs are no long within 90 meters of each other), authentication system **201-2** may automatically revoke access for the target UE to provide service or other communications to the requesting UE, and/or vice versa. In some embodiments, authentication system **201-2** may continue to monitor the locations of these UEs and may automatically reauthorize the requesting UE to receive service from or otherwise access the target UE (e.g., without the need for the requesting UE to initiate a subsequent request for access to the target UE).

(46) Data structure **401** may also include information reflecting particular UEs as administrators, blocked devices, etc. For example, as shown, UE **101-5** may be an administrator device, super-user device, etc. with respect to network **103-2**, inasmuch as UE **101-5** is authorized to access any target UEs of network **103-2** for any type of service with unlimited access. On the other hand, UE **101-6** may be a blocked device with respect to network **103-2**, inasmuch as UE **101-6** is not authorized to access (e.g., “no access”) any target UEs of network **103-2** for any type of service.

(47) Data structure **401** may also specify access policies associated with UEs of entire networks or other groups. For example, as shown data structure **401** may include an access policy that is applicable to all UEs of network **103-3**, for requests associated with a “AGV control” service directed to target UEs **101-2** and/or **101-4**. This access policy may specify that such requesting UEs may access the requested UEs and/or service for one hour, so long as the requesting UEs are within a particular region (represented as “Region_C”) at the time of the request and/or so long as the requesting UEs remain within the particular region while accessing the target UEs **101-2** and/or **101-4**. As discussed above, access for such requesting UEs may be revoked in instances where such requesting UEs move outside of the particular region. While some examples of access policies, as well as criteria based on which particular access policies may be determined as applicable to particular access requests, are discussed above, access policies maintained by policy repository **305** may include additional, fewer, different, and/or differently arranged access policies in accordance with some embodiments. For example, requesting and/or target UEs **101** may be specified based on

attributes, criteria, etc. of such UEs **101**. Such attributes, criteria, etc. may include a device type such (e.g., mobile phone, radio-controlled drone, semi-autonomous or remote-controlled robot, AGV, IoT device, etc.), physical attributes (e.g., screen size, device thickness, device weight, etc.), device capabilities (e.g., wireless bands or technologies supported, supported protocols or standards, etc.), labels or categories (e.g., “first responder,” “enterprise,” “mission critical,” etc.), and/or other attributes.

(48) Returning to FIG. 3, once authentication system **201-2** has identified (at **312**) the particular access policy that is applicable to access request **301**, authentication system **201-2** may determine (at **314**) whether to grant or deny the requested access of UE **101-2** by UE **101-1**. For example, authentication system **201-2** may make such determination based on the information regarding UE **101-1** (received at **308**) as well as the information regarding UE **101-2** (received at **310**).

Authentication system **201-2** may compare such information, associated with UEs **101-1** and/or **101-2**, to the identified (at **312**) access policy. In this example, authentication system **201-2** may determine, based on location information of UE **101-1** (received at **308**) that UE **101-1** is within Region_A, and may further determine, based on location information of UE **101-1** (received at **310**) that UE **101-2** is within Region_B. In this example, authentication system **201-2** may output (at **108**) a response to UE **101-1** that UE **101-1** is authorized for the requested access for two hours, in accordance with the identified access policy. Additionally, or alternatively, as discussed above, authentication system **201-2** may notify one or more other elements of network **103-2**, such as a gateway, a router, a firewall, target UE **101-2**, and/or one or more other devices or systems, that UE **101-1** is authorized to access (e.g., communicate with, receive service from, etc.) UE **101-2** for two hours. In some embodiments, authentication system **201-2** may generate an authentication token or other suitable authentication information and may provide such authentication information to requesting UE **101-1** and/or elements of network **103-2**. UE **101-1** and the respective network elements of network **103-2** may accordingly use such authentication and/or authorization information to facilitate the access by requesting UE **101-1** of target UE **101-2**. Additionally, or alternatively, such network elements of network **103-2** may be configured with an identifier of UE **101-1**, which may be used by the network elements to allow traffic to be sent to and/or received from (e.g., without blocking or dropping such traffic) UE **101-1**.

(49) FIG. 5 illustrates an example process **500** for a dynamic authentication and/or authorization of an external UE for access to a network. In some embodiments, some or all of process **500** may be performed by authentication system **201** (e.g., a first authentication system **201** of a first network **103-1**). In some embodiments, one or more other devices may perform some or all of process **500** in concert with, and/or in lieu of, authentication system **201**.

(50) As shown, process **500** may include monitoring (at **502**) information associated with one or more UEs **101** (e.g., a first UE **101**) that is associated with a first network **103**. For example, authentication system **201** of the first network **103** may receive, monitor, etc. UE information associated with UEs **101** that are associated with the first network **103** as a “home” network. Such information may be monitored, collected, etc. by devices or systems that are internal to and/or that otherwise implement the first network **103**, such as authentication system **201**, device integrity system **203**, location determination system **205**, security risk measurement system **209**, role-based directory system **211**, and/or some other suitable device or system. As discussed above, the UE information may include authentication information, device integrity information, location information, security risk measurement information, role/group information, and/or other suitable information.

(51) Process **500** may further include receiving (at **504**) an access request associated with the first UE **101** from a second UE **101** that is associated with a second network **103**. For example, authentication system **201** may receive access request **301** from the second UE **101** and/or some other source, and may identify that access request **301** is associated with the first UE **101**. For example, access request **301** may include an identifier of the first UE **101**, an identifier of a service

provided by the first UE **101**, and/or some other information based on which authentication system **201** may identify that the request is associated with the first UE **101**. In some embodiments, authentication system **201** may identify that the requesting UE **101** (e.g., the second UE **101**) is external, roaming, etc. with respect to the first network **103**. For example, authentication system **201** may identify that UDR **207** and/or some other element of the first network **103** does not include information indicating that the second UE **101** is registered with, native to, provisioned by, and/or otherwise associated with the first network **103** as a home network. Additionally, or alternatively, authentication system **201** may identify that UDR **207** and/or some other element of the first network **103** includes information indicating that the second UE **101** is not registered with, is not native to, and/or otherwise is not associated with the first network **103** as a home network (e.g., is a roaming UE, an external UE, etc. with respect to the first network **103**). In some embodiments access request **301** may include an identifier of the second network **103**, which is a home network of access request **301**.

(52) Process **500** may additionally include obtaining (at **506**) monitored information, associated with the first UE **101**, from the second network **103** (e.g., the home network **103** of UE **101**). For example, authentication system **201** may communicate with the second network **103** (e.g., with NEF **303** of the second network **103** and/or some other suitable device, system, interface, etc. of the second network **103**) to obtain the requested UE information associated with the second UE **101**. In some embodiments, the second network **103** may authenticate the second UE **101** and/or verify that the second UE **101** is authorized to receive the requested service prior to providing the requested UE information to authentication system **201** of the first network **103**. The authentication procedure may be performed based on a request from the first network **103** or performed periodically by the second network **103** using continuous authentication procedures dictated by policies associated with the second network **103**. As an example, a policy may dictate that the second network **103** re-authenticate the second UE **101** every 3 hours. Similarly, the policies of the first network **103** may require that the first network **103** re-authenticate the first UE **101** every 6 hours.

(53) Process **500** may also include identifying (at **508**) monitored information associated with the first UE **101**. For example, authentication system **201** may identify information monitored by one or more elements of the first network **103**, such as authentication and/or authorization information monitored or generated by authentication system **201**, device integrity information monitored or generated by device integrity system **203**, location information monitored or generated by location determination system **205**, security risk measurement information provided by security risk measurement system **209**, role/group information provided by role-based directory system **211**, and/or other suitable information. As discussed above, in some embodiments, authentication system **201** may receive such information from UDR **207** of the first network **103** and/or some other suitable device or system (e.g., directly from authentication system **201**, device integrity system **203**, location determination system **205**, security risk measurement system **209**, and/or role-based directory system **211**).

(54) Process **500** may further include identifying (at **510**) an access policy associated with the first UE **101** and the second UE **101**. For example, as discussed above, authentication system **201** may identify a set of access policies maintained by policy repository **305** of the first network **103**, which may include criteria, conditions, etc. indicating which particular policies are applicable to particular scenarios. The identified access policy may be an exact match, a “best” match, and/or may otherwise be identified based on comparing attributes of the first and/or second UEs **101** (e.g., identifiers, device groups, device types, etc.) to such criteria, conditions, etc.

(55) Process **500** may additionally include comparing (at **512**) the identified access policy to the monitored information associated with the first and second UEs **101** to determine whether to allow access to the first network **103** (e.g., to the first UE **101**). For example, the determination of whether to allow access may be based on the access request (received at **504**) or, as discussed below, a determination of whether to continue to allow access that has previously been granted

(e.g., based on continued monitoring (at **504** and/or **506**) of the first UE **101** and/or the second UE **101**). For example, authentication system **201** may compare location information, authentication information, device integrity information, security risk measurement information, role/group information, and/or other UE information associated with the first and second UEs **101** to conditions, criteria, restrictions, constraints, etc. specified in the identified (at **510**) access policy to determine whether to allow access.

(56) If, based on the comparing, authentication system **201** determines that access should not be granted and/or should be denied (at **514**—NO), then authentication system **201** may deny (at **516**) the request and/or may revoke previously granted (e.g., currently ongoing) access. For example, authentication system **201** may respond to the request (at **504**) with an indication that the requested access is not granted. Authentication system **201** may output an indication of why the requested access was not granted, such as an indication of which policy or policies would be violated (e.g., a location-based policy, an authentication-based policy, etc.). In some embodiments, the indication may include and/or may be represented by an error code, report, etc. In some embodiments, Authentication system **201** may perform some other action, such as activating an alarm, contacting law enforcement, adding one or more UEs **101** and/or users to a block list, etc.

(57) In some embodiments, the revocation (at **516**) may be a revocation of previously granted access, based on the ongoing monitoring of UEs **101** (e.g., at **504** and/or **506**). For example, the policy may have previously not been violated (e.g., UEs **101** may have been within a threshold distance of each other, as indicated by the policy) and access may have previously been granted for the second UE **101** to access the first UE **101**, but the first UE **101** and/or the second UE **101** may have moved, thus causing the policy to be violated. As another example, over time, the policy may have been changed, or authentication system **201** may have selected (at **510**) a different policy based on differing conditions. As discussed above, denying the request and/or revoking access may include communicating with the first UE **101** and/or one or more other elements of the first network **103** to indicate that the access of the second UE **101** has been denied or revoked, based on which the first UE **101** and/or other elements of the first network **103** may reject traffic, reject requests, and/or otherwise not provide service to the second UE **101**.

(58) If, on the other hand, authentication system **201** determines that access should be allowed (at **514**—YES), then process **500** may include accepting (at **518**) the request and/or reinstating access which was previously revoked, suspended, etc. For example, authentication system **201** may indicate, to the first and/or second UEs **101**, that the requested access has been granted. As discussed above, authentication system **201** may indicate, to one or more elements of the first network **103**, that the second UE **101** is authorized to communicate with the first UE **101** (e.g., to receive the requested service and/or to otherwise communicate with the first UE **101**). Further, in situations where the second UE **101** previously had access to communicate with the first UE **101**, but such access was revoked, suspended, etc., such access may be reinstated, restored, etc. In some embodiments, authentication system **201** may, in the absence of a previous request from the second UE **101**, maintain (at **518**) information indicating that the second UE **101** is eligible to access the first UE **101**. For example, authentication system **201** may “precheck” or “preauthorize” the second UE **101** to communicate with the first UE **101**, such that in the event that the second UE **101** eventually requests access to the first UE **101**, authentication system **201** may utilize such preauthorization information to determine that the second UE **101** is authorized to access the first UE **101**.

(59) FIG. **6** illustrates an example environment **600**, in which one or more embodiments may be implemented. In some embodiments, environment **600** may correspond to a Fifth Generation (“5G”) network, and/or may include elements of a 5G network. In some embodiments, environment **600** may correspond to a 5G Non-Standalone (“NSA”) architecture, in which a 5G radio access technology (“RAT”) may be used in conjunction with one or more other RATs (e.g., a Long-Term Evolution (“LTE”) RAT), and/or in which elements of a 5G core network may be

implemented by, may be communicatively coupled with, and/or may include elements of another type of core network (e.g., an evolved packet core (“EPC”)). In some embodiments, portions of environment **600** may represent or may include a 5G core (“5GC”). As shown, environment **600** may include UE **101**, RAN **610** (which may include one or more Next Generation Node Bs (“gNBs”) **611**), RAN **612** (which may include one or more evolved Node Bs (“eNBs”) **613**), and various network functions such as Access and Mobility Management Function (“AMF”) **615**, Mobility Management Entity (“MME”) **616**, Serving Gateway (“SGW”) **617**, Session Management Function (“SMF”)/Packet Data Network (“PDN”) Gateway (“PGW”)-Control plane function (“PGW-C”) **620**, Policy Control Function PCF/PCRF **625**, Application Function (“AF”) **630**, User Plane Function (“UPF”)/PGW-User plane function (“PGW-U”) **635**, Unified Data Management (“UDM”)/Home Subscriber Server (“HSS”) **640**, and AUSF **645**. Environment **600** may also include one or more networks, such as Data Network (“DN”) **650**. Environment **600** may include one or more additional devices or systems communicatively coupled to one or more networks (e.g., DN **650**), such as authentication system **201**, device integrity system **203**, location determination system **205**, UDR **207**, security risk measurement system **209**, role-based directory system **211**, NEF **303**, and/or policy repository **305**, which may perform one or more operations as discussed above.

(60) The example shown in FIG. **6** illustrates one instance of each network component or function (e.g., one instance of SMF/PGW-C **620**, PCF/PCRF **625**, UPF/PGW-U **635**, UDM/HSS **640**, and/or AUSF **645**). In practice, environment **600** may include multiple instances of such components or functions. For example, in some embodiments, environment **600** may include multiple “slices” of a core network, where each slice includes a discrete and/or logical set of network functions (e.g., one slice may include a first instance of SMF/PGW-C **620**, PCF/PCRF **625**, UPF/PGW-U **635**, UDM/HSS **640**, and/or AUSF **645**, while another slice may include a second instance of SMF/PGW-C **620**, PCF/PCRF **625**, UPF/PGW-U **635**, UDM/HSS **640**, and/or AUSF **645**). The different slices may provide differentiated levels of service, such as service in accordance with different Quality of Service (“QoS”) parameters.

(61) The quantity of devices and/or networks, illustrated in FIG. **6**, is provided for explanatory purposes only. In practice, environment **600** may include additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than illustrated in FIG. **6**. For example, while not shown, environment **600** may include devices that facilitate or enable communication between various components shown in environment **600**, such as routers, modems, gateways, switches, hubs, etc. In some implementations, one or more devices of environment **600** may be physically integrated in, and/or may be physically attached to, one or more other devices of environment **600**. Alternatively, or additionally, one or more of the devices of environment **600** may perform one or more network functions described as being performed by another one or more of the devices of environment **600**.

(62) Elements of environment **600** may interconnect with each other and/or other devices via wired connections, wireless connections, or a combination of wired and wireless connections. Examples of interfaces or communication pathways between the elements of environment **600**, as shown in FIG. **6**, may include an N1 interface, an N2 interface, an N3 interface, an N4 interface, an N5 interface, an N6 interface, an N7 interface, an N8 interface, an N9 interface, an N10 interface, an N11 interface, an N12 interface, an N13 interface, an N14 interface, an N26 interface, an S1-C interface, an S1-U interface, an S5-C interface, an S5-U interface, an S6a interface, an S11 interface, and/or one or more other interfaces. Such interfaces may include interfaces not explicitly shown in FIG. **6**, such as Service-Based Interfaces (“SBIs”), including an Namf interface, an Nudm interface, an Npcf interface, an Nupf interface, an Nnef interface, an Nsmf interface, and/or one or more other SBIs.

(63) In some embodiments, environment **600** may be, may include, may be implemented by, and/or may be communicatively coupled to a respective network **103**. For example, network **103-1** may

include one instance or implementation of environment **600**, and network **103-2** may include a different and/or separate instance or implementation of environment **600**. As another example, network **103-1** may include some or all of the elements of environment **600**, while network **103-2** may include different or differently arranged elements. In some embodiments, network **103-1** may include a first instance or implementation of authentication system **201**, device integrity system **203**, location determination system **205**, UDR **207**, and/or one or more other elements, while network **103-2** may include a different second instance or implementation of authentication system **201**, device integrity system **203**, location determination system **205**, UDR **207**, and/or one or more other elements.

(64) UE **101** may include a computation and communication device, such as a wireless mobile communication device that is capable of communicating with RAN **610**, RAN **612**, and/or DN **650**. UE **101** may be, or may include, a radiotelephone, a personal communications system (“PCS”) terminal (e.g., a device that combines a cellular radiotelephone with data processing and data communications capabilities), a personal digital assistant (“PDA”) (e.g., a device that may include a radiotelephone, a pager, Internet/intranet access, etc.), a smart phone, a laptop computer, a tablet computer, a camera, a personal gaming system, an IoT device (e.g., a sensor, a smart home appliance, a wearable device, a Machine-to-Machine (“M2M”) device, or the like), or another type of mobile computation and communication device. UE **101** may send traffic to and/or receive traffic (e.g., user plane traffic) from DN **650** via RAN **610**, RAN **612**, and/or UPF/PGW-U **635**.

(65) RAN **610** may be, or may include, a 5G RAN that includes one or more base stations (e.g., one or more gNBs **611**), via which UE **101** may communicate with one or more other elements of environment **600**. UE **101** may communicate with RAN **610** via an air interface (e.g., as provided by gNB **611**). For instance, RAN **610** may receive traffic (e.g., user plane traffic such as voice call traffic, data traffic, messaging traffic, etc.) from UE **101** via the air interface, and may communicate the traffic to UPF/PGW-U **635** and/or one or more other devices or networks.

Further, RAN **610** may receive signaling traffic, control plane traffic, etc. from UE **101** via the air interface, and may communicate such signaling traffic, control plane traffic, etc. to AMF **615** and/or one or more other devices or networks. Additionally, RAN **610** may receive traffic intended for UE **101** (e.g., from UPF/PGW-U **635**, AMF **615**, and/or one or more other devices or networks) and may communicate the traffic to UE **101** via the air interface.

(66) RAN **612** may be, or may include, a LTE RAN that includes one or more base stations (e.g., one or more eNBs **613**), via which UE **101** may communicate with one or more other elements of environment **600**. UE **101** may communicate with RAN **612** via an air interface (e.g., as provided by eNB **613**). For instance, RAN **612** may receive traffic (e.g., user plane traffic such as voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE **101** via the air interface, and may communicate the traffic to UPF/PGW-U **635** (e.g., via SGW **617**) and/or one or more other devices or networks. Further, RAN **612** may receive signaling traffic, control plane traffic, etc. from UE **101** via the air interface, and may communicate such signaling traffic, control plane traffic, etc. to MME **616** and/or one or more other devices or networks. Additionally, RAN **612** may receive traffic intended for UE **101** (e.g., from UPF/PGW-U **635**, MME **616**, SGW **617**, and/or one or more other devices or networks) and may communicate the traffic to UE **101** via the air interface.

(67) AMF **615** may include one or more devices, systems, Virtualized Network Functions (“VNFs”), Cloud-Native Network Functions (“CNFs”), etc., that perform operations to register UE **101** with the 5G network, to establish bearer channels associated with a session with UE **101**, to hand off UE **101** from the 5G network to another network, to hand off UE **101** from the other network to the 5G network, manage mobility of UE **101** between RANs **610** and/or gNBs **611**, and/or to perform other operations. In some embodiments, the 5G network may include multiple AMFs **615**, which communicate with each other via the N14 interface (denoted in FIG. **6** by the line marked “N14” originating and terminating at AMF **615**).

(68) MME **616** may include one or more devices, systems, VNFs, CNFs, etc., that perform operations to register UE **101** with the EPC, to establish bearer channels associated with a session with UE **101**, to hand off UE **101** from the EPC to another network, to hand off UE **101** from another network to the EPC, manage mobility of UE **101** between RANs **612** and/or eNBs **613**, and/or to perform other operations.

(69) SGW **617** may include one or more devices, systems, VNFs, CNFs, etc., that aggregate traffic received from one or more eNBs **613** and send the aggregated traffic to an external network or device via UPF/PGW-U **635**. Additionally, SGW **617** may aggregate traffic received from one or more UPF/PGW-U **635** and may send the aggregated traffic to one or more eNBs **613**. SGW **617** may operate as an anchor for the user plane during inter-eNB handovers and as an anchor for mobility between different telecommunication networks or RANs (e.g., RANs **610** and **612**).

(70) SMF/PGW-C **620** may include one or more devices, systems, VNFs, CNFs, etc., that gather, process, store, and/or provide information in a manner described herein. SMF/PGW-C **620** may, for example, facilitate the establishment of communication sessions on behalf of UE **101**. In some embodiments, the establishment of communications sessions may be performed in accordance with one or more policies provided by PCF/PCRF **625**.

(71) PCF/PCRF **625** may include one or more devices, systems, VNFs, CNFs, etc., that aggregate information to and from the 5G network and/or other sources. PCF/PCRF **625** may receive information regarding policies and/or subscriptions from one or more sources, such as subscriber databases and/or from one or more users (such as, for example, an administrator associated with PCF/PCRF **625**).

(72) AF **630** may include one or more devices, systems, VNFs, CNFs, etc., that receive, store, and/or provide information that may be used in determining parameters (e.g., quality of service parameters, charging parameters, or the like) for certain applications.

(73) UPF/PGW-U **635** may include one or more devices, systems, VNFs, CNFs, etc., that receive, store, and/or provide data (e.g., user plane data). For example, UPF/PGW-U **635** may receive user plane data (e.g., voice call traffic, data traffic, etc.), destined for UE **101**, from DN **650**, and may forward the user plane data toward UE **101** (e.g., via RAN **610**, SMF/PGW-C **620**, and/or one or more other devices). In some embodiments, multiple UPFs **635** may be deployed (e.g., in different geographical locations), and the delivery of content to UE **101** may be coordinated via the N9 interface (e.g., as denoted in FIG. **6** by the line marked “N9” originating and terminating at UPF/PGW-U **635**). Similarly, UPF/PGW-U **635** may receive traffic from UE **101** (e.g., via RAN **610**, RAN **612**, SMF/PGW-C **620**, and/or one or more other devices), and may forward the traffic toward DN **650**. In some embodiments, UPF/PGW-U **635** may communicate (e.g., via the N4 interface) with SMF/PGW-C **620**, regarding user plane data processed by UPF/PGW-U **635**.

(74) UDM/HSS **640** and AUSF **645** may include one or more devices, systems, VNFs, CNFs, etc., that manage, update, and/or store, in one or more memory devices associated with AUSF **645** and/or UDM/HSS **640**, profile information associated with a subscriber. AUSF **645** and/or UDM/HSS **640** may perform authentication, authorization, and/or accounting operations associated with the subscriber and/or a communication session with UE **101**.

(75) UDR **207** may include one or more devices, systems, VNFs, CNFs, etc., that, as discussed above, receive information from authentication system **201**, device integrity system **203**, location determination system **205**, security risk measurement system **209**, and/or role-based directory system **211**. In some embodiments, UDR **207** and/or one or more other devices or systems may receive information (e.g., associated with one or more UEs **101** that are associated with a network implemented by environment **600** as a home network) from one or more other sources. For example, UDR **207** and/or policy repository **305** may receive policy information from PCF/PCRF **625** and/or some other source. In some embodiments, UDR **207** and/or authentication system **201** may receive UE authentication information from UDM/HSS **640**, UE **101**, and/or some other source. As another example, UDR **207** and/or device integrity system **203** may receive device

integrity information from UDM/HSS **640**, UE **101**, and/or some other source. In some embodiments, UDR **207** and/or location determination system **205** may receive UE location information from AMF **615**, MME **616**, UE **101**, and/or some other suitable device or system. In some embodiments, UDR **207** may receive risk scores computed and/or otherwise provided by security risk measurement system **209**. Similarly, UDR **207** may receive role and/or group information associated with one or more UEs **101** from a directory that is updated regularly when a role associated with such UEs **101** change, such as role-based directory system **211**.

(76) NEF **303** may include one or more devices, systems, VNFs, CNFs, etc., that perform one or more operations discussed above. For example, NEF **303** may maintain authentication and/or authorization information indicating devices or systems, external to a particular network **103** implemented by environment **600**, that are authorized to receive information associated with such network **103**. Further, NEF **303** may maintain authorization information indicating types or other attributes, constraints, limits, etc. on information that is authorized to be provided to such external device or system. In some embodiments, NEF **303** may receive UE information, as discussed above, from UDR **207** and/or one or more other sources, and may provide some or all such information to an authorized device or system (e.g., a particular authentication system **201** of some other network **103**).

(77) DN **650** may include one or more wired and/or wireless networks. For example, DN **650** may include an Internet Protocol IP-based PDN, a wide area network (“WAN”) such as the Internet, a private enterprise network, and/or one or more other networks. UE **101** may communicate, through DN **650**, with data servers, other UEs **101**, and/or to other servers or applications that are coupled to DN **650**. DN **650** may be connected to one or more other networks, such as a public switched telephone network (“PSTN”), a public land mobile network (“PLMN”), and/or another network. DN **650** may be connected to one or more devices, such as content providers, applications, web servers, and/or other devices, with which UE **101** may communicate.

(78) FIG. 7 illustrates an example RAN environment **700**, which may be included in and/or implemented by one or more RANs (e.g., RAN **610**, RAN **612**, or some other RAN). In some embodiments, a particular RAN may include one RAN environment **700**. In some embodiments, a particular RAN may include multiple RAN environments **700**. In some embodiments, RAN environment **700** may correspond to a particular gNB **611** of a 5G RAN (e.g., RAN **610**). In some embodiments, RAN environment **700** may correspond to multiple gNBs **611**. In some embodiments, RAN environment **700** may correspond to one or more other types of base stations of one or more other types of RANs. As shown, RAN environment **700** may include Central Unit (“CU”) **705**, one or more Distributed Units (“DUs”) **703-1** through **703-N** (referred to individually as “DU **703**,” or collectively as “DUs **703**”), and one or more Radio Units (“RUs”) **701-1** through **701-M** (referred to individually as “RU **701**,” or collectively as “RUs **701**”).

(79) CU **705** may communicate with a core of a wireless network (e.g., may communicate with one or more of the devices or systems described above with respect to FIG. 6, such as AMF **615** and/or UPF/PGW-U **635**). In the uplink direction (e.g., for traffic from UEs **101** to a core network), CU **705** may aggregate traffic from DUs **703**, and forward the aggregated traffic to the core network. In some embodiments, CU **705** may receive traffic according to a given protocol (e.g., Radio Link Control (“RLC”)) from DUs **703**, and may perform higher-layer processing (e.g., may aggregate/process RLC packets and generate Packet Data Convergence Protocol (“PDCP”) packets based on the RLC packets) on the traffic received from DUs **703**.

(80) In accordance with some embodiments, CU **705** may receive downlink traffic (e.g., traffic from the core network) for a particular UE **101**, and may determine which DU(s) **703** should receive the downlink traffic. DU **703** may include one or more devices that transmit traffic between a core network (e.g., via CU **705**) and UE **101** (e.g., via a respective RU **701**). DU **703** may, for example, receive traffic from RU **701** at a first layer (e.g., physical (“PHY”) layer traffic, or lower PHY layer traffic), and may process/aggregate the traffic to a second layer (e.g., upper PHY and/or

RLC). DU **703** may receive traffic from CU **705** at the second layer, may process the traffic to the first layer, and provide the processed traffic to a respective RU **701** for transmission to UE **101**. (81) RU **701** may include hardware circuitry (e.g., one or more RF transceivers, antennas, radios, and/or other suitable hardware) to communicate wirelessly (e.g., via an RF interface) with one or more UEs **101**, one or more other DUs **703** (e.g., via RUs **701** associated with DUs **703**), and/or any other suitable type of device. In the uplink direction, RU **701** may receive traffic from UE **101** and/or another DU **703** via the RF interface and may provide the traffic to DU **703**. In the downlink direction, RU **701** may receive traffic from DU **703**, and may provide the traffic to UE **101** and/or another DU **703**.

(82) One or more elements of RAN environment **700** may, in some embodiments, be communicatively coupled to one or more Multi-Access/Mobile Edge Computing (“MEC”) devices, referred to sometimes herein simply as “MECs” **707**. For example, DU **703-1** may be communicatively coupled to MEC **707-1**, DU **703-N** may be communicatively coupled to MEC **707-N**, CU **705** may be communicatively coupled to MEC **707-2**, and so on. MECs **707** may include hardware resources (e.g., configurable or provisionable hardware resources) that may be configured to provide services and/or otherwise process traffic to and/or from UE **101**, via a respective RU **701**.

(83) For example, DU **703-1** may route some traffic, from UE **101**, to MEC **707-1** instead of to a core network via CU **705**. MEC **707-1** may process the traffic, perform one or more computations based on the received traffic, and may provide traffic to UE **101** via RU **701-1**. In some embodiments, MEC **707** may include, and/or may implement, some or all of the functionality described above with respect to AF **630**, UPF **635**, and/or one or more other devices, systems, VNFs, CNFs, etc. In this manner, ultra-low latency services may be provided to UE **101**, as traffic does not need to traverse DU **703**, CU **705**, links between DU **703** and CU **705**, and an intervening backhaul network between RAN environment **700** and the core network.

(84) FIG. **8** illustrates example components of device **800**. One or more of the devices described above may include one or more devices **800**. Device **800** may include bus **810**, processor **820**, memory **830**, input component **840**, output component **850**, and communication interface **860**. In another implementation, device **800** may include additional, fewer, different, or differently arranged components.

(85) Bus **810** may include one or more communication paths that permit communication among the components of device **800**. Processor **820** may include a processor, microprocessor, or processing logic that may interpret and execute instructions. In some embodiments, processor **820** may be or may include one or more hardware processors. Memory **830** may include any type of dynamic storage device that may store information and instructions for execution by processor **820**, and/or any type of non-volatile storage device that may store information for use by processor **820**.

(86) Input component **840** may include a mechanism that permits an operator to input information to device **800** and/or other receives or detects input from a source external to input component **840**, such as a touchpad, a touchscreen, a keyboard, a keypad, a button, a switch, a microphone or other audio input component, etc. In some embodiments, input component **840** may include, or may be communicatively coupled to, one or more sensors, such as a motion sensor (e.g., which may be or may include a gyroscope, accelerometer, or the like), a location sensor (e.g., a Global Positioning System (“GPS”)-based location sensor or some other suitable type of location sensor or location determination component), a thermometer, a barometer, and/or some other type of sensor. Output component **850** may include a mechanism that outputs information to the operator, such as a display, a speaker, one or more light emitting diodes (“LEDs”), etc.

(87) Communication interface **860** may include any transceiver-like mechanism that enables device **800** to communicate with other devices and/or systems. For example, communication interface **860** may include an Ethernet interface, an optical interface, a coaxial interface, or the like.

Communication interface **860** may include a wireless communication device, such as an infrared

("IR") receiver, a Bluetooth® radio, or the like. The wireless communication device may be coupled to an external device, such as a remote control, a wireless keyboard, a mobile telephone, etc. In some embodiments, device **800** may include more than one communication interface **860**. For instance, device **800** may include an optical interface and an Ethernet interface.

(88) Device **800** may perform certain operations relating to one or more processes described above. Device **800** may perform these operations in response to processor **820** executing software instructions stored in a computer-readable medium, such as memory **830**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may include space within a single physical memory device or spread across multiple physical memory devices. The software instructions may be read into memory **830** from another computer-readable medium or from another device. The software instructions stored in memory **830** may cause processor **820** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(89) The foregoing description of implementations provides illustration and description, but is not intended to be exhaustive or to limit the possible implementations to the precise form disclosed. Modifications and variations are possible in light of the above disclosure or may be acquired from practice of the implementations.

(90) For example, while series of blocks and/or signals have been described above (e.g., with regard to FIGS. 1-5), the order of the blocks and/or signals may be modified in other implementations. Further, non-dependent blocks and/or signals may be performed in parallel. Additionally, while the figures have been described in the context of particular devices performing particular acts, in practice, one or more other devices may perform some or all of these acts in lieu of, or in addition to, the above-mentioned devices.

(91) The actual software code or specialized control hardware used to implement an embodiment is not limiting of the embodiment. Thus, the operation and behavior of the embodiment has been described without reference to the specific software code, it being understood that software and control hardware may be designed based on the description herein.

(92) In the preceding specification, various example embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

(93) Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of the possible implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one other claim, the disclosure of the possible implementations includes each dependent claim in combination with every other claim in the claim set.

(94) Further, while certain connections or devices are shown, in practice, additional, fewer, or different, connections or devices may be used. Furthermore, while various devices and networks are shown separately, in practice, the functionality of multiple devices may be performed by a single device, or the functionality of one device may be performed by multiple devices. Further, multiple ones of the illustrated networks may be included in a single network, or a particular network may include multiple networks. Further, while some devices are shown as communicating with a network, some such devices may be incorporated, in whole or in part, as a part of the network.

(95) To the extent the aforementioned implementations collect, store, or employ personal

information of individuals, groups or other entities, it should be understood that such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage, and use of such information can be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as can be appropriate for the situation and type of information. Storage and use of personal information can be in an appropriately secure manner reflective of the type of information, for example, through various access control, encryption and anonymization techniques for particularly sensitive information.

(96) No element, act, or instruction used in the present application should be construed as critical or essential unless explicitly described as such. An instance of the use of the term “and,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Similarly, an instance of the use of the term “or,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Also, as used herein, the article “a” is intended to include one or more items, and may be used interchangeably with the phrase “one or more.” Where only one item is intended, the terms “one,” “single,” “only,” or similar language is used. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

Claims

1. A device, comprising: one or more processors configured to: monitor information associated with a set of User Equipment (“UEs”) associated with a first network; receive an access request, for access to a first UE of the set of UEs, from a second UE associated with a second network; obtain, based on the access request and from the second network, monitored information associated with the second UE; identify monitored information associated with the first UE; identify a particular access policy that is associated with the first UE and the second UE; identify, based on the particular access policy, the monitored information associated with the first UE, and the monitored information associated with the second UE, whether to grant or deny the access request; and output, in response to the access request, an indication of whether the access request is granted or denied.
2. The device of claim 1, wherein the monitored information associated with the first UE is determined by one or more devices of the first network that communicate with the first UE.
3. The device of claim 1, wherein the monitored information associated with the first UE includes location information associated with the first UE, and wherein the monitored information associated with the second UE includes location information associated with the second UE.
4. The device of claim 1, wherein the first network is a home network of the first UE, and wherein the second network is a home network of the second UE.
5. The device of claim 4, wherein the access request indicates that the second network is the home network of the second UE, wherein the one or more processors are further configured to: communicate with the second network to obtain the monitored information associated with the second UE based on the indication that the second network is the home network of the second UE.
6. The device of claim 1, wherein the second network is a private network that utilizes wireless network infrastructure provided by the first network.
7. The device of claim 1, wherein the access request includes a service type, wherein identifying the particular access policy further includes identifying that the particular access policy is associated with the requested service type.
8. A non-transitory computer-readable medium, storing a plurality of processor-executable instructions to: monitor information associated with a set of User Equipment (“UEs”) associated with a first network; receive an access request, for access to a first UE of the set of UEs, from a second UE associated with a second network; obtain, based on the access request and from the

second network, monitored information associated with the second UE; identify monitored information associated with the first UE; identify a particular access policy that is associated with the first UE and the second UE; identify, based on the particular access policy, the monitored information associated with the first UE, and the monitored information associated with the second UE, whether to grant or deny the access request; and output, in response to the access request, an indication of whether the access request is granted or denied.

9. The non-transitory computer-readable medium of claim 8, wherein the monitored information associated with the first UE is determined by one or more devices of the first network that communicate with the first UE.

10. The non-transitory computer-readable medium of claim 8, wherein the monitored information associated with the first UE includes location information associated with the first UE, and wherein the monitored information associated with the second UE includes location information associated with the second UE.

11. The non-transitory computer-readable medium of claim 8, wherein the first network is a home network of the first UE, and wherein the second network is a home network of the second UE.

12. The non-transitory computer-readable medium of claim 11, wherein the access request indicates that the second network is the home network of the second UE, wherein the plurality of processor-executable instructions further include processor-executable instructions to: communicate with the second network to obtain the monitored information associated with the second UE based on the indication that the second network is the home network of the second UE.

13. The non-transitory computer-readable medium of claim 8, wherein the second network is a private network that utilizes wireless network infrastructure provided by the first network.

14. The non-transitory computer-readable medium of claim 8, wherein the access request includes a service type, wherein identifying the particular access policy further includes identifying that the particular access policy is associated with the requested service type.

15. A method, comprising: monitoring information associated with a set of User Equipment (“UEs”) associated with a first network; receiving an access request, for access to a first UE of the set of UEs, from a second UE associated with a second network; obtaining, based on the access request and from the second network, monitored information associated with the second UE; identifying monitored information associated with the first UE; identifying a particular access policy that is associated with the first UE and the second UE; identifying, based on the particular access policy, the monitored information associated with the first UE, and the monitored information associated with the second UE, whether to grant or deny the access request; and outputting, in response to the access request, an indication of whether the access request is granted or denied.

16. The method of claim 15, wherein the monitored information associated with the first UE is determined by one or more devices of the first network that communicate with the first UE.

17. The method of claim 15, wherein the monitored information associated with the first UE includes location information associated with the first UE, and wherein the monitored information associated with the second UE includes location information associated with the second UE.

18. The method of claim 15, wherein the first network is a home network of the first UE, and wherein the second network is a home network of the second UE, wherein the access request indicates that the second network is the home network of the second UE, the method further comprising: communicating with the second network to obtain the monitored information associated with the second UE based on the indication that the second network is the home network of the second UE.

19. The method of claim 15, wherein the second network is a private network that utilizes wireless network infrastructure provided by the first network.

20. The method of claim 15, wherein the access request includes a service type, wherein identifying

the particular access policy further includes identifying that the particular access policy is associated with the requested service type.
